

Unique Paper Code	:	32341303
Name of the Course	:	B.Sc. (H) Computer Science
Name of the Paper	:	Computer Networks
Semester	:	III
Duration	:	3 Hours
Maximum Marks	:	75
Year of Admission	:	2019

Instructions for Candidates:

Attempt any Four out of Six questions.

All Questions carry equal marks.

- Q1** What are the three flow control/Error recovery protocols used for noisy channel? Compare and contrast the Go-Back-N ARQ with Selective Repeat ARQ. Consider that a 1Mbps satellite link connects two ground stations. The altitude of the satellite is 36,504 km and speed of the signal is 3×10^8 m/s. Assume that the acknowledgment packets are negligible in size and that there are no errors during communication. What should be the packet size for a channel utilization of 25% for the satellite link using go-back-127 sliding window protocol? Further consider that a message $M = 1010001101$ needs to be transmitted over this link. Cyclic redundancy check (CRC) with divisor polynomial $x^5 + x^4 + x^2 + 1$ is used for error detection. What will be the final message transmitted over the link?
- Q2** Explain the circuit switching and packet switching techniques. How is virtual circuit approach different from datagram approach? A message of size 500 bytes has to be sent in a network having bandwidth 10Mbps. Assume that each packet contains a header of 100 bytes and packet switching technique is used. If the message is divided into 5 packets, what would be the transmission delay?
- Q3** What characteristics of the carrier signal are changed to represent the digital signal in each of the following modulation techniques: - ASK, FSK, PSK and QAM? Which of the four digital to analog modulation techniques (ASK, FSK, PSK, QAM) is most susceptible to noise? Defend your answer.
In a QAM modulation scheme, there are 4 amplitude levels and 16 phase levels and bit rate is 72 Kbps. Calculate the baud rate.

Q4 The following character encoding is used in a data link protocol:

A: 11010101; B: 10101001; FLAG: 01111110; ESC: 10100011.

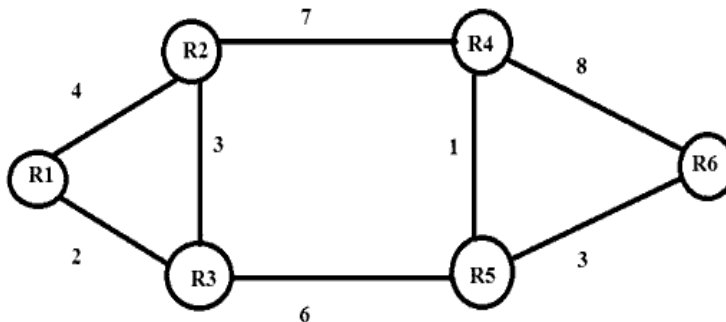
Show the bit sequence transmitted (in binary) for the five character frame. A ESC B ESC FLAG when each of the following framing methods are used:

- i. **Character count**
- ii. **Flag bytes with byte stuffing.**
- iii. **Starting and ending flag bytes, with bit stuffing**

Also find the original data for the given output obtained after applying byte-stuffing technique:

FLAG A B ESC ESC C ESC ESC ESC FLAG ESC FLAG D E FLAG.

Q5 What is Count to infinity problem? Explain it in brief with suitable example. Consider a network with 6 routers R1 to R6 connected with links having weights as shown in the following diagram.



All the routers use the **distance vector routing algorithm** to update their routing tables. Each router starts with its routing table initialized to contain an entry for each neighbor with the weight of the respective connecting link. After all the routing tables stabilize, how many links in the network will never be used for carrying any data? Also show the distance vector maintained at each router.

Q6 The network **200.1.2.30** has been subdivided into 4 subnets.

- (i) Which class the given IP address belongs to?
- (ii) Give the subnet mask for the given IP address.
- (iii) Give the IP addresses of these 4 subnets.
- (iv) How many hosts can be on each subnet?
- (v) Determine the starting IP address and the last IP address of each subnet.
- (vi) Determine which network the IP address **200.1.2.130** belongs to.
- (vii) Determine the limited broadcast address for each subnet.

Unique Paper Code	:	32341303-OC
Name of the Course	:	B.Sc. (H) Computer Science
Name of the Paper	:	Computer Networks
Semester	:	III
Duration	:	3 Hours
Maximum Marks	:	75
Year of Admission	:	2015, 2016, 2017, 2018

Instructions for Candidates:

**Attempt any Four out of Six questions.
All Questions carry equal marks.**

- Q1** The network **200.1.2.30** has been subdivided into 4 subnets.
- (i) Which class the given IP address belongs to?
 - (ii) Give the subnet mask for the given IP address.
 - (iii) Give the IP addresses of these 4 subnets.
 - (iv) How many hosts can be on each subnet?
 - (v) Determine the starting IP address and the last IP address of each subnet.
 - (vi) Determine which network the IP address **200.1.2.130** belongs to.
 - (vii) Determine the limited broadcast address for each subnet.
- Q2** What characteristics of the carrier signal are changed to represent the digital signal in each of the following modulation techniques: - ASK, FSK, PSK and QAM?
Which of the four digital to analog modulation techniques (ASK, FSK, PSK, QAM) is most susceptible to noise? Defend your answer.
In a QAM modulation scheme, there are 4 amplitude levels and 16 phase levels and bit rate is 72 Kbps. Calculate the baud rate.
- Q3** The following character encoding is used in a data link protocol:
A: 11010101; B: 10101001; FLAG: 01111110; ESC: 10100011.
Show the bit sequence transmitted (in binary) for the five character frame. A ESC B ESC FLAG when each of the following framing methods are used:
- i. **Character count**
 - ii. **Flag bytes with byte stuffing.**
 - iii. **Starting and ending flag bytes, with bit stuffing**

Also find the original data for the given output obtained after applying byte-stuffing technique:

FLAG A B ESC ESC C ESC ESC ESC FLAG ESC FLAG D E FLAG.

Q4 Draw the pulse diagram for the bit stream **0011001100110011** and also calculate the average signal rate for the following line coding schemes:

- i. **Manchester**
- ii. **Differential Manchester**
- iii. **Polar Return to Zero**
- iv. **Polar Non Return to Zero - Level**
- v. **Polar Non Return to Zero – Inversion**

Explain the problems associated with Polar NRZ techniques.

Q5 What is the **maximum window size** for data transmission in **Selective Repeat and Go Back N protocol** if n bit frame sequence numbers are used? Consider a scenario for **Selective Repeat protocol** where frames with sequence numbers from 0 to 4 have been transmitted by the sender. Now assume that the following sequence of event occurs:

- (i) frame 0 does time-out
- (ii) A new frame with sequence number 5 is transmitted.
- (iii) frame 1 and frame 2 time-out
- (iv) A new frame with sequence number 6 is transmitted.

At every stage show the status of the sender's window i.e. the outstanding frames in the sender's window.

Consider another scenario for **Go Back N sliding window protocol** with window size 3 where sender X needs to send a message consisting of 9 packets to receiver Y. If every 5th packet that sender X transmits gets lost (but no ACKs from receiver Y ever get lost), then how many packets sender X will transmit for sending the message to receiver Y?

Q6 How long will it take a station to realize that there has been a collision in CSMA/CD protocol? Show the working of Back-off algorithm for the following scenario:

Assume X and Y are the only two stations on the Ethernet. Each of them has a long queue of frames to be transmitted. Both X and Y attempts to transmit a frame at the same time and the frames collide. Assume that X wins the first Back-off race. At the end of this successful transmission by X, both X and Y attempt to transmit again and collide. Show all the possible cases and find the probability of station X wins over station Y.